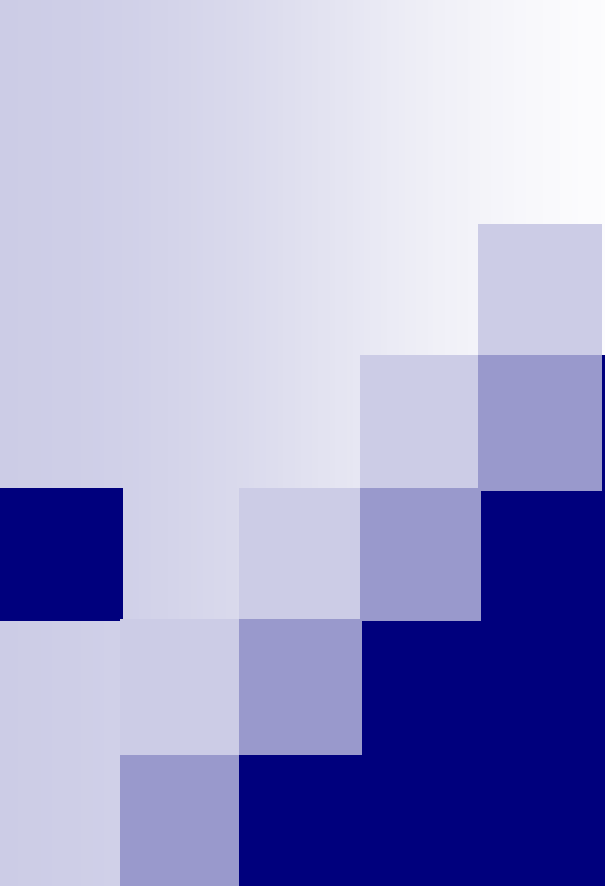
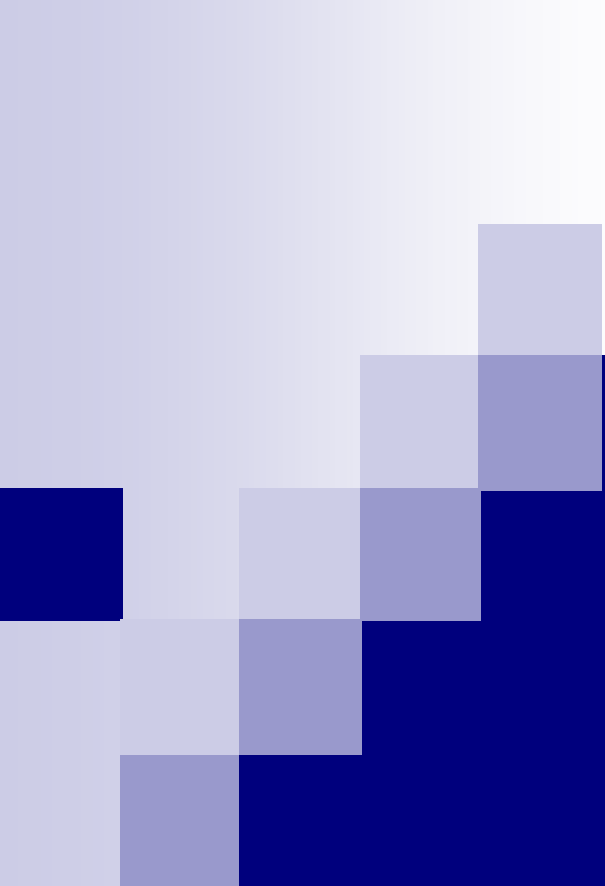




SCM Plan and SQA Plan



Lecture # 37



Software Configuration Management Plan



IEEE SCM Plan

- Introduction
- Reference Documents, Definitions, Acronyms
- Management
- Activities
- Resources
- Plan Maintenance
- Plan Approval



1. Introduction - 1

- Introduction information provides a simplified overview of the configuration management activities so that those approving, performing, or interacting with the Software Configuration Management Plan can obtain a clear understanding of the SCM Plan.

Introduction - 2

- Explains the purpose and content of the Configuration Management Plan (i.e., the methodical storage and recording of all software components and deliverables during development)
- The SCM Plan documents methods to be used for the identification of software items, control and implementation of change, and recording and reporting change implementation status

Introduction - 3

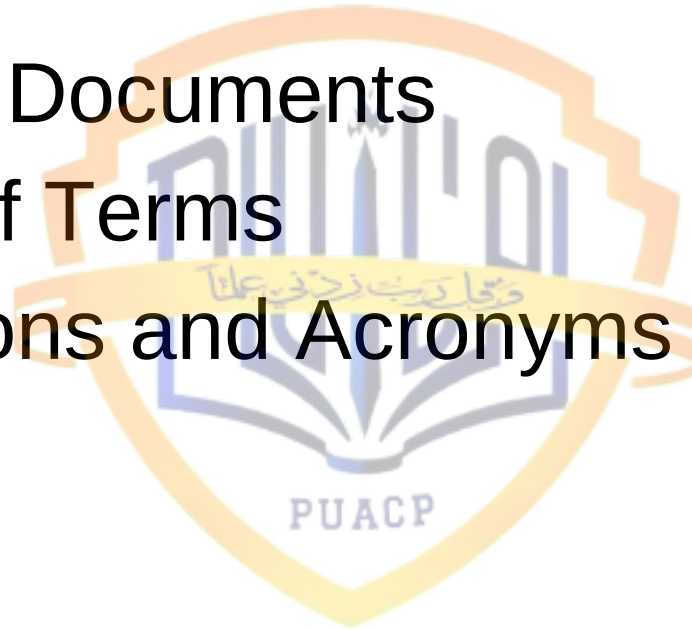
- The plan should describe methods used for:
 - Identification of software configuration items,
 - Control and implementation of change
 - Recording and reporting change and problem report implementation status
 - Conducting configuration audits
 - Review and approval cycles as well as approval authority, and
 - Identification of personnel responsible for configuration management

2. Reference Documents, Definitions, Acronyms - 1

- This section provides a complete list of documents referenced elsewhere in the text of the SCM Plan. By definition, these documents originate outside the project. Also included in this section is a glossary of project specific terms and their definitions and a list of project-specific abbreviations and acronyms and their meaning

Reference Documents, Definitions, Acronyms - 2

- Reference Documents
- Glossary of Terms
- Abbreviations and Acronyms



3. Management - 1

- This section provides information describing the allocation of responsibilities and authorities for software configuration management activities to organizations and individuals within the project structure
- Organization
 - This section depicts the organizational context, both technical and managerial, within which the prescribed software configuration management activities are to be implemented

Management - 2

- Responsibilities

- Describes the allocation of software configuration management activities to organizational units

- Policies, Directives and Procedures

- Any external constraints, or requirements, placed on the SCM Plan by other policies, directives, or procedures must be identified here. A detailed impact analysis should accompany the identification of external constraints

4. Activities - 1

- Identifies all functions and tasks required to manage the configuration of the software system as specified in the scope of the SCM Plan. Both technical and managerial activities must be identified

Activities - 2

■ Configuration Identification

- Identify, name, and describe the documented physical and functional characteristics of the code, specification, design, and data elements to be controlled for the project. The Plan must identify the items to be maintained in configuration management control
- Identifying configuration items
- Naming configuration items
- Acquiring configuration items

Activities - 3

■ Configuration Control

- Configuration control activities request, evaluate, approve or disapprove, and implement changes to the software configuration items. Changes include both error correction and enhancement. This section shall identify the records to be used for tracking and documenting the sequence of steps for each change

Activities - 4

- Configuration Control (cont'd)
 - Requesting changes
 - Evaluating changes
 - Approving or disapproving changes
 - Implementing changes

Activities - 5

■ Configuration Status Accounting

- Record and report the status of configuration items. The following minimum data elements should be tracked and reported for each configuration management item:
 - Approved version
 - Status of requested changes
 - Implementation status of approved changes

Activities - 6

■ Configuration Audits and Reviews

- Configuration audits determine the extent to which the actual configuration management items reflect the required physical and functional characteristics. Configuration reviews may also be designed a management tool used to ensure that a software configuration management baseline is established

Activities - 7

■ Interface Control

- Coordinates changes to the project's configuration management items with changes to interfacing items outside the scope of the SCM Plan

Activities - 8

■ Subcontractor/Vendor Control

- For acquired software, the Software Configuration Management Plan shall describe how the vendor software will be received, tested, and placed under software configuration management control
- For both subcontracted and acquired software, the SCM Plan must define the activities to be performed to incorporate the externally developed items into project configuration management and to coordinate changes to these items

5. Resources - 1

- Establishes the sequence and coordination for all the software configuration management activities and all the events affecting the Plan's implementation

Resources - 2

■ Schedules

- Schedule information shall be expressed as absolute dates, as dates relative to other project activities, as project milestones, or as a simple sequence of events. Graphic representation can be particularly appropriate for conveying this information

■ Resources

- Identifies the software tools, techniques, equipment, personnel, and training necessary for the implementation of software configuration management activities

6. Plan Maintenance

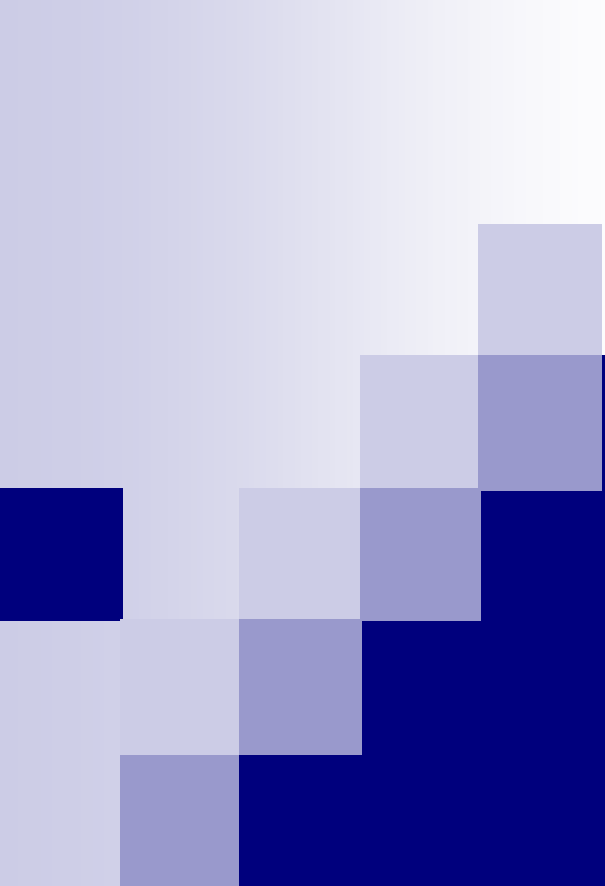
- Identifies and describes the activities and responsibilities necessary to ensure continued software configuration management planning during the life cycle of the project. This section of the SCM Plan should state the following:
 - Who is responsible for monitoring the SCM Plan
 - How frequently updates are to be applied
 - How changes to the SCM Plan are to be evaluated and approved
 - How changes to the SCM Plan are to be made and communicated

7. Plan Approvals



References for SCM Plan

- SCM Plan based on IEEE Standard for Software Configuration Management Plans (Std 828-1990) and the IEEE Guide to Software Configuration Management (Std 1042-1987)
- Managing the Software Process by Watts S. Process, AW, 1989, (Chapter 12.1) [pp.228-232]



Software Quality Assurance Plan (SQAP)

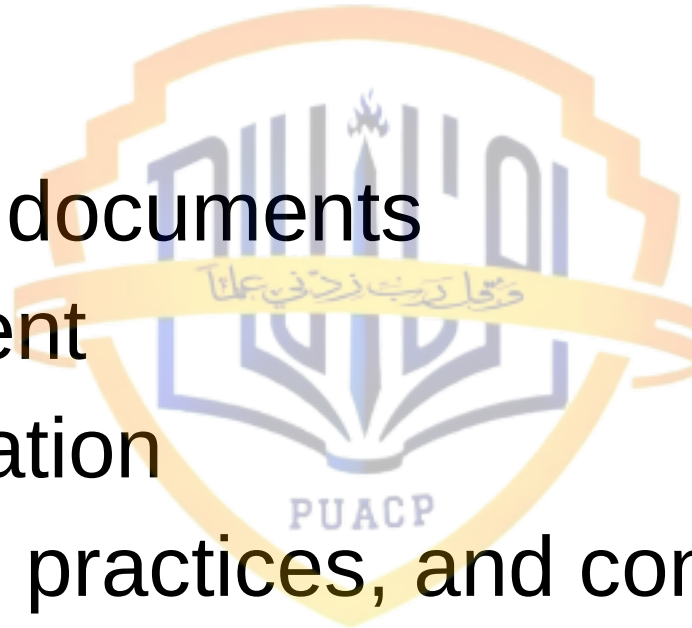


SQA Plan

- Every development and maintenance project should have a software quality assurance plan (SQAP) that specifies:
 - Its goals
 - The SQA tasks to be performed
 - The standards against which the development work is to be measured
 - The procedures and organizational structure

IEEE SQAP Standard - 1

- Purpose
- Reference documents
- Management
- Documentation
- Standards, practices, and conventions
- Reviews and audits
- Software configuration management



IEEE SQAP Standard - 2

- Test
- Problem reporting and corrective action
- Tools, techniques, and methodologies
- Media control
- Supplier control
- Records collection, maintenance, and retention

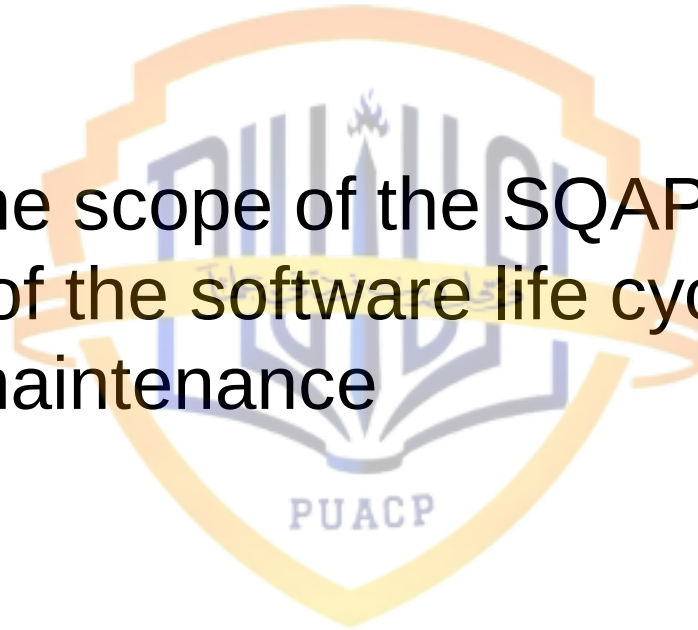
1. Purpose - 1

- This section documents the purpose of this Software Quality Assurance (SQA) Plan
- It documents the goals, processes, and responsibilities required to implement effective quality assurance functions for the current project

Purpose - 2

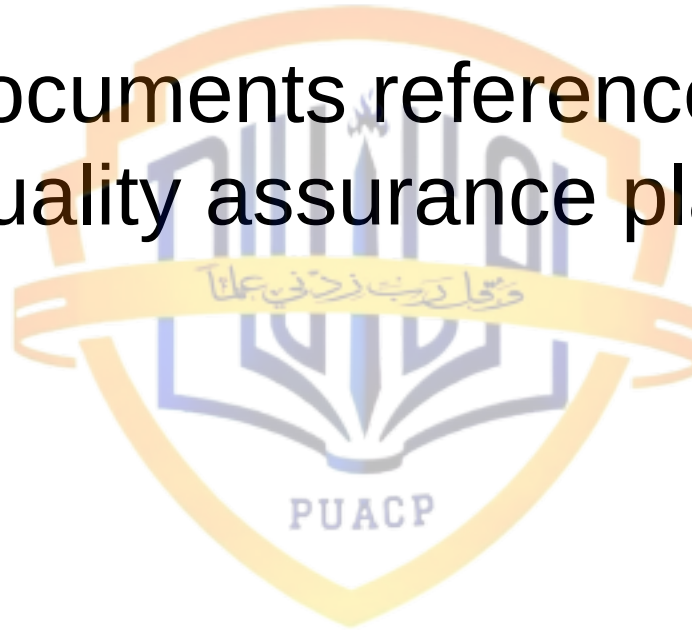
- Scope

- Defines the scope of the SQAP in different activities of the software life cycle and even through maintenance



2. Reference Documents

- Lists the documents referenced in this software quality assurance plan



3. Management - 1

- This section describes the management organizational structure, its roles and responsibilities, and the software quality tasks to be performed
- Management organization
- Tasks
- Roles and responsibilities
- Software assurance estimated resources

Management - 2

- Management Organization
 - Describes the support of entities, organizations and personnel
 - Relevant entities/roles that are of interest and applicable to this SQA Plan and the software assurance effort specially include
 - Project office
 - Assurance management office

Management - 3

■ Tasks

- This section summarizes the tasks (product and process assessments) to be performed during the development, operations, and maintenance of software
- These tasks are selected based on the developer's Project Schedule, Software Management Plan (SMP) (and/or Software Maintenance Plan) and planned deliverables, contractual deliverables, and identified reviews

Management - 4

- Tasks (cont'd)
 - Product assessments
 - Peer Review packages
 - Document Reviews
 - Software Development Folders
 - Software Configuration Management
 - Test results
 - Process assessments

Management - 5

■ Tasks (cont'd)

□ Process assessments

- Project Planning
- Project Monitoring and Control
- Measurement and Analysis
- System/Subsystem Reviews
- Peer Reviews
- Requirements Management
- Software Configuration Management and Configuration Audits
- Test Management (Verification & Validation)
- Software Problem Reporting and Corrective Action
- Risk Management
- Supplier Agreement Management

Management - 6

- Roles and Responsibilities
 - This section describes the roles and responsibilities for each assurance person assigned to a project
- Software Assurance Manager
- Software Quality Personnel

Management - 7

- Software Assurance Estimated Resources
 - Staffing to support software assurance (i.e., quality, safety, and reliability) activities must be balanced against various project characteristics and constraints, including cost, schedule, maturity level of the providers, criticality of the software being developed, return on investment, perceived risk, etc.

4. Documentation - 1

- This section identifies the minimum documentation governing the requirements, development, verification, validation, and maintenance of software that falls within the scope of this software quality plan

Documentation - 2

- Each document below shall be assessed (reviewed) by SQ personnel
 - The software requirements specifications
 - The software design description
 - The software verification and validation report
 - User documentation

5. Standards, Practices, and Conventions

- This section highlights the standards, practices, quality requirements, and metrics to be applied to ensure a successful software quality program
- This section specifies a minimum content of:
 - Documentation standards
 - Logic structure standards
 - Coding standards
 - Commentary standards
- Software Quality Program
- Standard Metrics

6. Reviews and Audits - 1

- This section discusses major project reviews conducted by SQA staff and software team members
- **Generic review guidelines**
 - A set of guidelines for all formal technical reviews or inspections is presented in this section
 - Conducting a review
 - General guidelines for conducting a review

Reviews and Audits - 2

- Roles and responsibilities
 - The roles people play during a FTR or inspection and the responsibilities of each player
- Review work products
 - Documents, forms, lists produced as a consequence of the FTR/inspection
- Formal technical reviews/inspections
 - A description of the specific character and the intent of each major FTR/inspection conducted during the software process

Reviews and Audits - 3

■ SQA audits

- A description of audits performed by the SQA group with the intent of assessing how well SQA and software engineering activities are being conducted on a project

7. Software Configuration Management

- A brief overview of the content of the software configuration management (SCM) plan is presented here
- Alternatively, the SCM plan is referenced

8. Test

- SQA personnel will assure that the test management processes and products are being implemented per the Software Management Plan and /or Test Plans
- This includes all types of testing of software system components as described in the test plan, specifically during integration testing (verification) and acceptance testing (validation).

9. Problem Reporting and Corrective Action - 1

- This section describes problem reporting mechanisms that occur as a consequence of the formal technical reviews' or inspections' that are conducted and the means for corrective action and follow-up
- Reporting mechanisms
 - Describes how and to whom problems are reported

Problem Reporting and Corrective Action - 2

- Responsibilities

- ☐ Describes who has responsibility for corrective actions and follow-up

- Data collection and evaluation

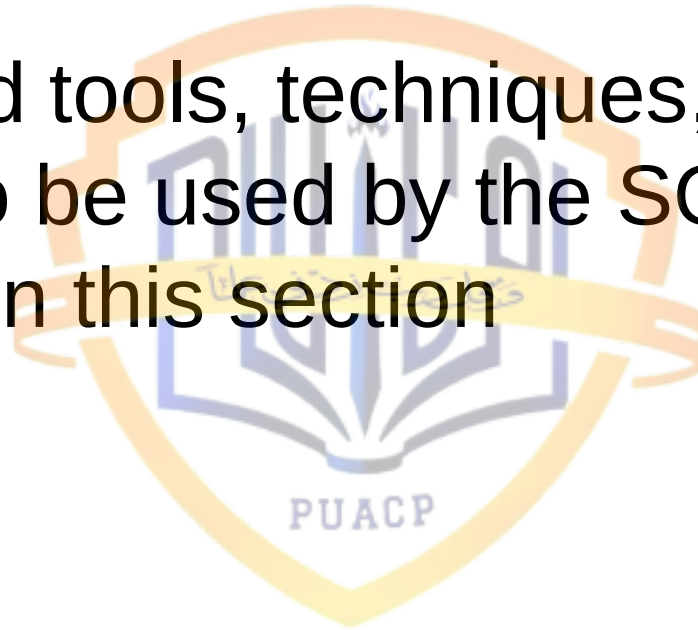
- ☐ Describes the manner in which error/defect data are collected and stored for future or real-time evaluation

Problem Reporting and Corrective Action - 3

- Statistical SQA
 - Describes the quantitative techniques that will be applied to error/defect data in an effort to discern trends and improvement

10. Tools, Techniques, and Methodologies

- Specialized tools, techniques, and methods to be used by the SQA group are described in this section



11. Media Control



12. Supplier Control

- SQA personnel will conduct off-site surveillance activities at supplier sites on software development activities
- SQA personnel will conduct a baseline assessment of the suppliers' Quality Management Systems (QMS) to ensure that the suppliers' have quality processes in place. This initial assessment will help to scope the level of effort and follow-on activities in the area of software quality assurance

13. Record Collection, Maintenance, and Retention

- SQA personnel will maintain records that document assessments performed on the project. Maintaining these records will provide objective evidence and traceability of assessments performed throughout the project's life cycle
- Example records include the process and product assessments reports, completed checklists, the SQA Activity Schedule, metrics, weekly status reports, etc.

References for SQA Plan

- Managing the Software Process by Watts S. Process, AW, 1989, (Chapter 8.4) [pp.147-150]
- SQA Plan Template by R.S. Pressman & Associates, Inc., <http://www.rspa.com/docs/sqaplan.html>
- IEEE template for SQA Plan